

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

by

TAEHONG KWON

Student number: 28576810

submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

SUPERVISOR: Prof. Ernest Mnkandla

CO-SUPERVISOR: Prof. Donatien Koulla Moulla

CO-SUPERVISOR: Dr. David Sena Attipoe

July 2026

4 Implementation and Empirical Results

Chapter 3 specified how EndorseRank and the Adaptive Weighted PageRank (AWP) baseline are built, which construct checks are applied, how uncertainty is quantified and how runtime is timed. This chapter applies that design to Arbitrum One and reports what was observed. It is organised as a single case study: one chain, one observation window, one matched cohort for all same-window comparisons, one spender cohort for the temporal holdout and one expanded address pool for node-count sensitivity. Tables are generated from the evaluation summary in Section 3.13, and confidence intervals accompany every Kendall τ coefficient.

4.1 Case study setting: Arbitrum One

Arbitrum One was selected as the case because it is a high-activity Ethereum rollup with mature DeFi protocols (Section 2.2), its logs are available in the public BigQuery export used by the pipeline, and a margin-trading protocol on it (GMX V2) supplies the wallet seed set described below. The observation window is 1 December 2025 to 31 May 2026. Six months is long enough to span several market regimes and to give tenure and active-month proxies room to vary, and short enough that a latest-allowance snapshot remains contemporaneous with the transfer history that AWP aggregates.

4.1.1 Cohorts

Three address sets are used, each for one purpose.

- Matched cohort ($n = 5,521$). Wallets seeded from GMX V2 margin-trading activity that have non-zero connectivity in both the latest-allowance endorsement graph G_{approve} and the time-decayed transfer graph G_{transfer} over the window. All same-window alignment, difference-test

and benchmark results refer to this cohort. Requiring connectivity in both graphs ensures that neither method is evaluated on wallets for which it can only return the teleportation floor.

- Holdout spender cohort ($n = 1,335$). Wallets that appear as spenders in at least one allowance during the scoring period (events up to 28 February 2026). Their scores are tested against owner–spender approval pairs and transfer senders first observed between 1 March and 31 May 2026 (Section 3.7).
- Expanded wallet pool ($N = 99,080$). The matched cohort unioned with 93,559 supplemental addresses sampled from monthly ERC-20 activity on the chain. The pool is used only for node-count sensitivity of runtime and for the sample-definition sensitivity analysis (Section 3.12.1); its limitation, that supplemental addresses contribute nodes but no additional extracted edges, is restated where the results are reported.

Graph sizes for the matched cohort are given with the benchmark in Table 4.1. The endorsement graph has roughly one ninth as many edges as the transfer graph on the same wallets. That asymmetry is a property of the two representations rather than of the extraction: both graphs are built from the same wallet seed set and window, but EndorseRank keeps only the latest decimal-adjusted allowance per owner–spender pair, so repeated approvals collapse to one directed edge, whereas AWP accumulates every transfer between two wallets into a decayed weight. Differences in scores and alignment come from ranking logic and graph semantics rather than from inconsistent sampling.

4.1.2 Implementation

All methods run through the same Python PageRank solver with identical damping ($d = 0.85$), tolerance (10^{-8}) and iteration cap; the single-layer methods differ only in their edge lists, and the coupled operator mixes the two lists per node as in Equation (3.6) (Section 3.15). Scores, per-proxy statistics, bootstrap intervals and timings are written to one summary file per run; the tables below were produced from the run of 14 September 2026. The repository revision, the configuration and the summary files for that run are recorded in Appendix 8.1.

4.2 Computational benchmark (Claim C1)

Claim C1 asserts that EndorseRank is computationally cheaper than AWP on identical inputs. Table 4.1 reports PageRank solve time, its standard deviation over the five timed repeats, peak memory during the warm-up solve, iterations to convergence and the size of each method’s evaluation subgraph on the matched cohort, following the timing protocol of Section 3.11.

Table 4.1: Computational benchmark on the matched wallet cohort ($n = 5,521$; AWP/EndorseRank runtime ratio $10.2\times$). Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs. $|V|$, $|E|$ = nodes and edges of each method’s evaluation subgraph. C-PR walks on the union of both layers, so $|E|$ is the sum of the two edge sets; S-PR runs on the transfer layer and its runtime includes the EndorseRank solve that supplies its restart vector.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$ V $	$ E $
EndorseRank	0.401	0.016	4.6	37.0	6,442	14,727
AWP	4.105	0.193	38.2	100.0	30,791	137,087
C-PR ($\lambda = 0.5$)	4.760	0.471	41.3	76.0	31,242	151,814
S-PR	4.156	0.757	39.1	145.0	30,791	137,087

Three mechanisms account for the runtime ratio in the table. First, both parts of the timed solve, assembling the sparse transition operator from the edge list and each power-iteration step, are linear in $|E|$ (Appendix 8.3.3), and the endorsement graph has about one ninth of the transfer graph’s edges. Second, EndorseRank converges in roughly a third of the iterations AWP needs; the transfer operator, with its long-tailed decayed weights, mixes more slowly. Third, peak memory follows $|E|$ as well, since adjacency, weight and residual buffers scale with the edge list. The standard deviations show that run-to-run noise is an order of magnitude smaller than the gap between methods.

The two hybrid rows show what the speed result does and does not cover. C-PR walks on the union of the two edge sets (151,814 edges, 31,242 nodes), and its solve takes 4.76 s, slightly longer than AWP’s; it converges in fewer iterations than AWP (76 against 100), but each iteration touches more edges. S-PR runs on the transfer layer and takes 4.16 s including the EndorseRank solve that supplies its restart vector; it needs 145 iterations, since a concentrated restart distribution slows

the approach to the stationary vector. Claim C1 is therefore a statement about EndorseRank alone. Any method that reads the transfer layer, whether on its own or mixed with allowances, pays the transfer layer’s cost.

Two remarks bound the claim. The benchmark times the solver call on pre-built edge lists; end-to-end refresh cost includes extraction, decoding and edge-list construction, which are shared or partially shared across methods and are not part of C1. And “faster” is not “better”: the runtime result acquires meaning only when paired with the alignment results of Sections 4.3–4.4, which is why Claim C4 combines the two.

4.2.1 Node-count sensitivity on the expanded pool

Table 4.2 reports runtime at three staged node counts drawn from the expanded pool by deterministic SHA256 subsampling, together with the per-stage $|V|$ and $|E|$ of each method’s subgraph. In-cohort stages ($n \leq 5,521$) are given in Appendix 8.1 (Table 8.1).

Table 4.2: Runtime versus evaluation-set size on the expanded wallet pool ($N = 99,080$; SHA256 subsampling seed benchmark-tier2-v1). Each stage keeps the edges that touch at least one sampled wallet, so $|V|$ and $|E|$ are reported per stage; the pool adds addresses but no additional extracted edges beyond the matched-cohort subgraph, hence the full-pool row reproduces the matched-cohort graph. Speedup = AWP/ER runtime ratio within the same row. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
10,000	0.070	0.854	12.2×	1,296	2,055	8,489	26,732
50,000	0.334	2.706	8.1×	4,440	9,190	19,990	92,887
99,080	0.401	4.105	10.2×	6,442	14,727	30,791	137,087

Two features of the table need to be read together with Section 3.12.1. First, the full-pool row is numerically identical to the matched-cohort row of Table 4.1. This is not a copy: the supplemental addresses have no extracted approval or transfer logs, so the induced subgraph at $N = 99,080$ contains exactly the cohort edges, and the timing protocol times each distinct graph once per session and reuses that measurement wherever the same graph is tabulated. The same measurement therefore appears in the $d = 0.85$ row of Table 4.8. Second, $|E|$ is not monotone in n across

stages, because each stage is the subgraph induced by whichever wallets the hash order selects; the $n = 10,000$ stage happens to select comparatively few cohort wallets and hence few edges. What the table supports is a bounded statement: across stages whose edge counts vary by a factor of five to seven, the AWP/EndorseRank runtime ratio stays within roughly $8\text{--}12\times$ and EndorseRank remains below half a second. It does not show how either method scales when the edge count itself grows with the address population, since the second-tier extraction that would have supplied those edges did not complete.

Figure 4.1 plots the in-cohort and pool stages on a common log axis.

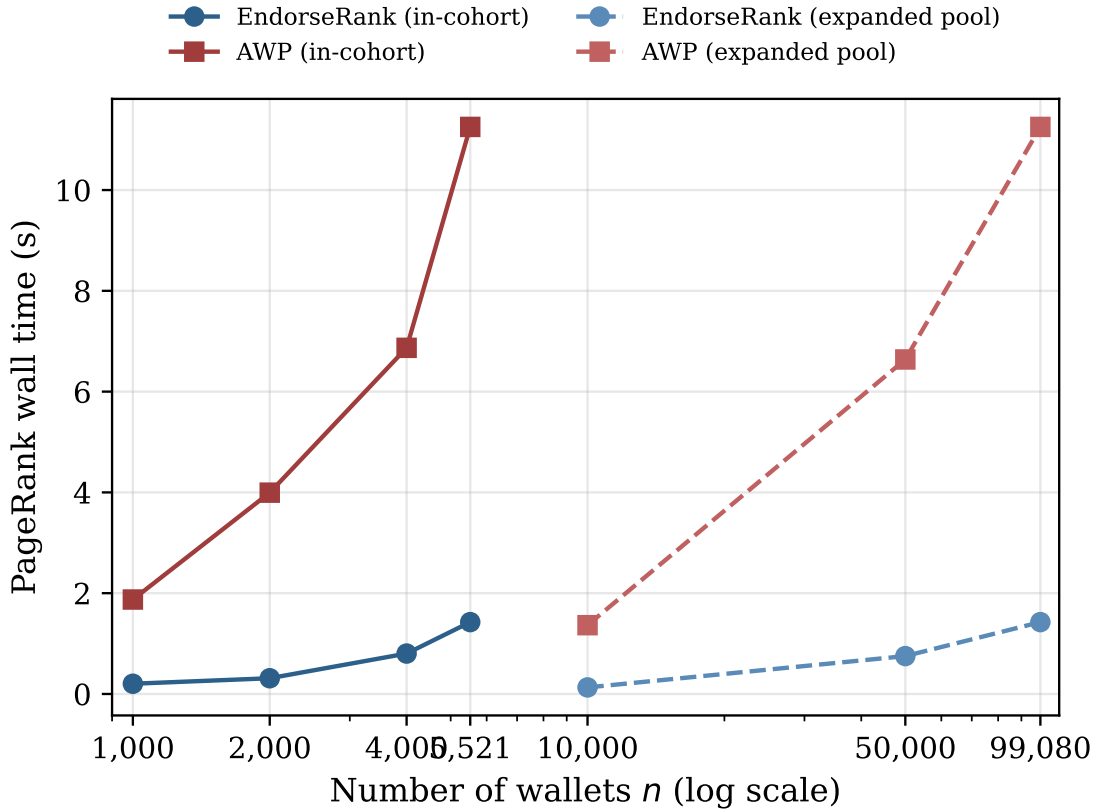


Figure 4.1: PageRank wall-clock runtime versus number of sampled wallets (log x -axis). Solid lines: in-cohort stages ($n \leq 5,521$, Table 8.1). Dashed lines: expanded-pool stages ($N = 99,080$, Table 4.2). Pool stages add addresses without adding extracted edges, so the rightmost pool point coincides with the full matched cohort.

4.3 Same-window alignment: EndorseRank vs. AWP (Claims C2–C3)

Claims C2 and C3 ask whether each method recovers its intended construct on contemporaneous local statistics and whether the winners are construct-specific. Table 4.3 reports family-mean Kendall τ with 95% paired-bootstrap intervals for the transfer, allowance and Sybil-stability families, together with the inter-method agreement between the two score vectors. Figure 4.2 shows the same family means as a heatmap.

Table 4.3: Family-mean Kendall τ with bootstrap confidence intervals on the matched cohort ($n = 5,521$). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Family	ER mean τ	ER 95% CI	AWP mean τ	AWP 95% CI
Transfer	0.333	[0.314, 0.353]	0.534	[0.522, 0.547]
Allowance	0.355	[0.322, 0.384]	-0.030	[-0.045, -0.017]
Sybil	0.233	[0.214, 0.253]	0.286	[0.274, 0.300]
EndorseRank vs. AWP score agreement: $\tau = 0.316$, CI [0.298, 0.336]				

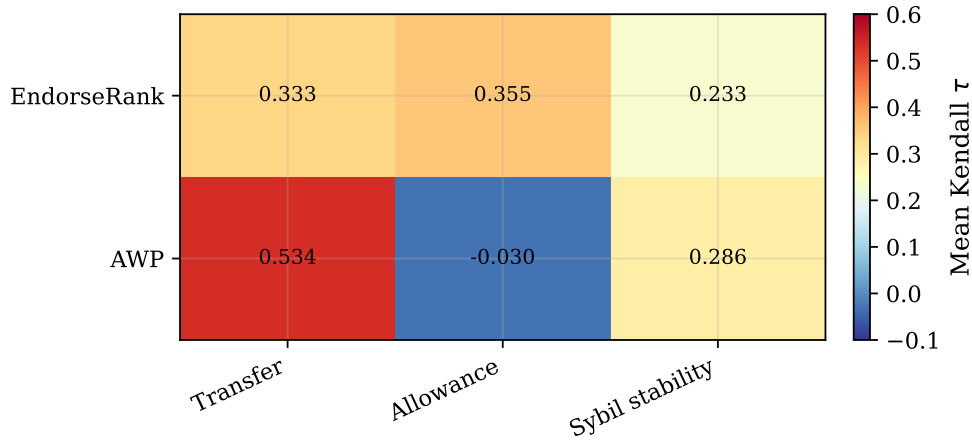


Figure 4.2: Family-mean Kendall τ on the matched cohort ($n = 5,521$) for EndorseRank and AWP; the values are those of Table 4.3.

The pattern is crossed. AWP aligns most strongly with the transfer family and is slightly negative on the allowance family; EndorseRank aligns most strongly with the allowance family and moderately with the transfer family. Whether each contrast exceeds sampling variation is tested in

Table 4.4, which reports the five pre-specified single-layer paired differences of Section 3.10; the hybrid contrasts are in Table 4.12 (Section 4.7).

Table 4.4: Paired bootstrap tests of differences between Kendall τ coefficients (matched cohort, $n = 5,521$). Family entries use the family-mean τ ; the same wallet resamples are used for a and b , so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.355	0.333	+0.022	[-0.026, 0.067]	0.810
AWP transfer minus EndorseRank transfer	0.534	0.333	+0.201	[0.182, 0.218]	1.000
EndorseRank allowance minus AWP allowance	0.355	-0.030	+0.385	[0.350, 0.421]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.286	0.233	+0.052	[0.038, 0.065]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.356	0.344	+0.012	[-0.041, 0.060]	0.675

Three of the five contrasts have intervals that exclude zero: AWP’s transfer alignment exceeds EndorseRank’s, EndorseRank’s allowance alignment exceeds AWP’s, and AWP exceeds EndorseRank on the Sybil-stability family. Two do not. EndorseRank’s allowance alignment is not statistically higher than its own transfer alignment (contrast i), and EndorseRank’s in-approve-degree alignment is not statistically higher than its in-degree alignment (contrast v). These two null results bound the interpretation of the allowance family. EndorseRank is a global smoothing of spender in-approve degree, so its agreement with allowance proxies is an intended-construct check, not an external test. Transfer centrality (AWP) carries no information about who receives spending authorisation, whereas endorsement centrality (EndorseRank) does, and the reverse holds for transfer volume. The within-method claim that EndorseRank prefers allowance over transfer signal is not supported.

Claim C2 is supported in the sense that each method achieves non-trivial alignment on its own construct with an interval well away from zero. Claim C3 is supported by the two between-method contrasts (ii and iii) whose intervals exclude zero in opposite directions: neither method dominates both checks.

4.4 Proxy family detail

The family means summarise per-proxy coefficients. This section reports Spearman ρ , Kendall τ and the 95% interval of τ for each proxy. Proxies are oriented so that higher values imply higher expected reputation before correlation, and the transfer proxies are those used to validate AWP in its source paper^[3].

4.4.1 Transfer family

Table 4.5: Domain alignment with transfer proxies (inbound in-degree and in-value, the validation proxies used for the AWP baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-degree	0.422	0.344	[0.324, 0.366]
In-value (sum inbound)	0.398	0.321	[0.303, 0.342]
<i>AWP</i>			
In-degree	0.791	0.617	[0.604, 0.631]
In-value (sum inbound)	0.610	0.450	[0.436, 0.467]

AWP’s agreement with in-degree is the largest coefficient in the chapter, and its agreement with in-value is smaller but still far above EndorseRank’s. Both are primary-construct checks for AWP, since it ranks on the time-decayed transfer graph; the informative observation is that AWP does not collapse to in-degree: logistic decay and value weighting move a measurable share of pairwise orderings. EndorseRank’s transfer coefficients are moderate and their intervals overlap neither zero nor AWP’s intervals. Wallets that receive many transfers are somewhat more likely to be endorsed spenders, but the two behaviours are far from interchangeable. If a protocol’s notion of reputation is inbound economic flow, AWP is the appropriate method on this evidence.

Table 4.6: Domain alignment with allowance proxies (spender-side in-approve degree and value). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-approve degree	0.360	0.356	[0.323, 0.385]
In-approve value (sum)	0.360	0.354	[0.321, 0.383]
<i>AWP</i>			
In-approve degree	-0.037	-0.030	[-0.045, -0.017]
In-approve value (sum)	-0.037	-0.030	[-0.045, -0.017]

4.4.2 Allowance family

EndorseRank’s coefficients on in-approve degree and in-approve value are nearly equal, which indicates that on this cohort being approved by many owners and being approved for large aggregate value convey similar ranking information under latest-allowance semantics. AWP’s coefficients are slightly negative with intervals that exclude zero: transfer centrality does not recover authorisation. That dissociation is the main same-window finding. If transfer-based reputation were a sufficient proxy for endorsement, AWP would show at least moderate allowance alignment; it does not, so protocols that care about which addresses are trusted to move tokens on behalf of others cannot substitute AWP for an allowance-aware method. The caveat of Section 4.3 applies: EndorseRank’s own allowance coefficients are a construct check, and the out-of-window test of the same construct is the holdout of Section 4.8.

4.4.3 Sybil-adjusted stability family

AWP exceeds EndorseRank on tenure and active months, as expected from time-decayed transfer edges that reward sustained activity; the paired difference on the family mean (Table 4.4, contrast iv) excludes zero. The inbound-counterparty ratio is weak for both methods and negative for AWP. This family is a robustness note on how much longitudinal transfer information each method carries, not a headline claim.

Table 4.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
Inbound counterparty ratio	0.093	0.073	[0.046, 0.097]
Transfer tenure (days)	0.366	0.297	[0.276, 0.318]
Active months	0.381	0.330	[0.310, 0.351]
<i>AWP</i>			
Inbound counterparty ratio	-0.063	-0.061	[-0.080, -0.040]
Transfer tenure (days)	0.602	0.436	[0.421, 0.452]
Active months	0.622	0.482	[0.466, 0.499]

4.5 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 ask whether EndorseRank and AWP order the same wallets differently. The inter-method Kendall τ between the two score vectors, with its interval, is given in the last row of Table 4.3. The rankings are related but far from interchangeable: agreement exceeds chance because economically active wallets participate in both approval and transfer markets, yet a large fraction of pairwise orderings disagree.

Figure 4.3 shows the marginal score distributions. Both are right-skewed, as PageRank on sparse directed graphs typically is, but they differ in shape and in the number of wallets that sit at the teleportation floor. Skewness alone does not establish divergence, so Figure 4.4 plots the ranks directly: the cloud is positively sloped but wide, with substantial off-diagonal mass.

Rank divergence is the structural counterpart of construct-specific alignment. If the two methods produced nearly identical orderings, the family-level winners of Table 4.3 could be noise around a shared ranking; the observed divergence, together with the opposite-signed between-method contrasts of Table 4.4, supports reading the two as related but distinct instruments. H1 is supported. Operationally, a protocol that currently uses transfer-based scores cannot assume that switching to, or blending in, allowance-based scores will preserve its wallet ordering; the largest disagreements are expected among transfer hubs that are rarely endorsed as spenders and among endorsed

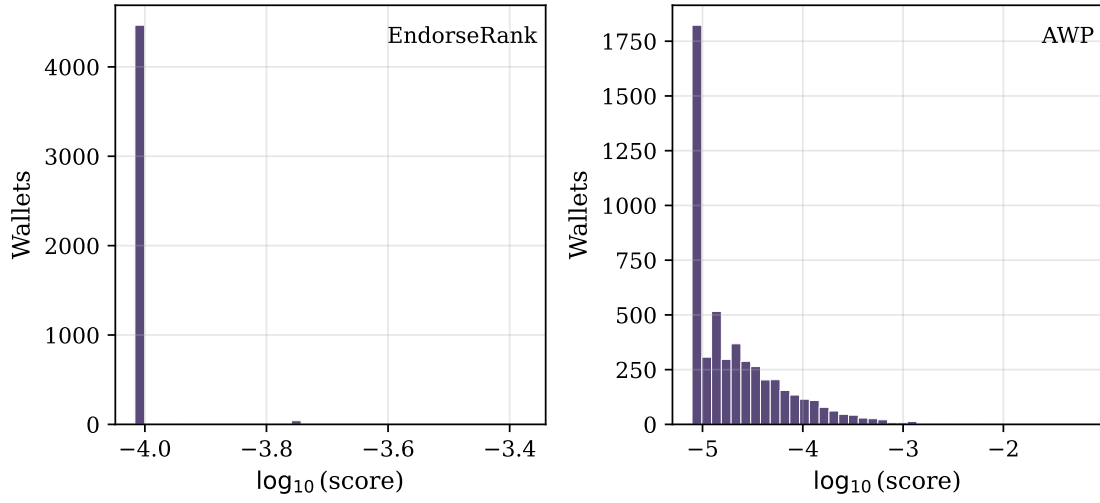


Figure 4.3: Marginal distributions of EndorseRank and AWP scores on the matched cohort ($n = 5,521$), on a $\log_{10}$ scale for wallets with positive score (EndorseRank has 966 zero-score wallets and AWP 381, omitted from the histograms).

spenders with small transfer footprints.

4.6 Robustness and sensitivity checks

The results above would be of limited use if they depended on a single damping value, a single token mix or a single sample definition. Three checks follow. The first two are robustness checks in the strict sense: the sample is held fixed and a parameter or the token universe is perturbed. The third is a sensitivity analysis: the sample definition itself is changed, and the coefficients are expected to move.

4.6.1 Damping

Family means are flat across $d \in \{0.75, 0.85, 0.95\}$ to the third decimal for EndorseRank and move by at most a few thousandths for AWP; the crossed pattern is not an artefact of the conventional $d = 0.85$. Because allowance edges encode a snapshot rather than a long path structure, changing how much mass teleports does not reorder wallets enough to alter family-level alignment. Runtimes do vary with d . AWP is markedly slowest at $d = 0.95$, consistent with the slower mixing of a high-

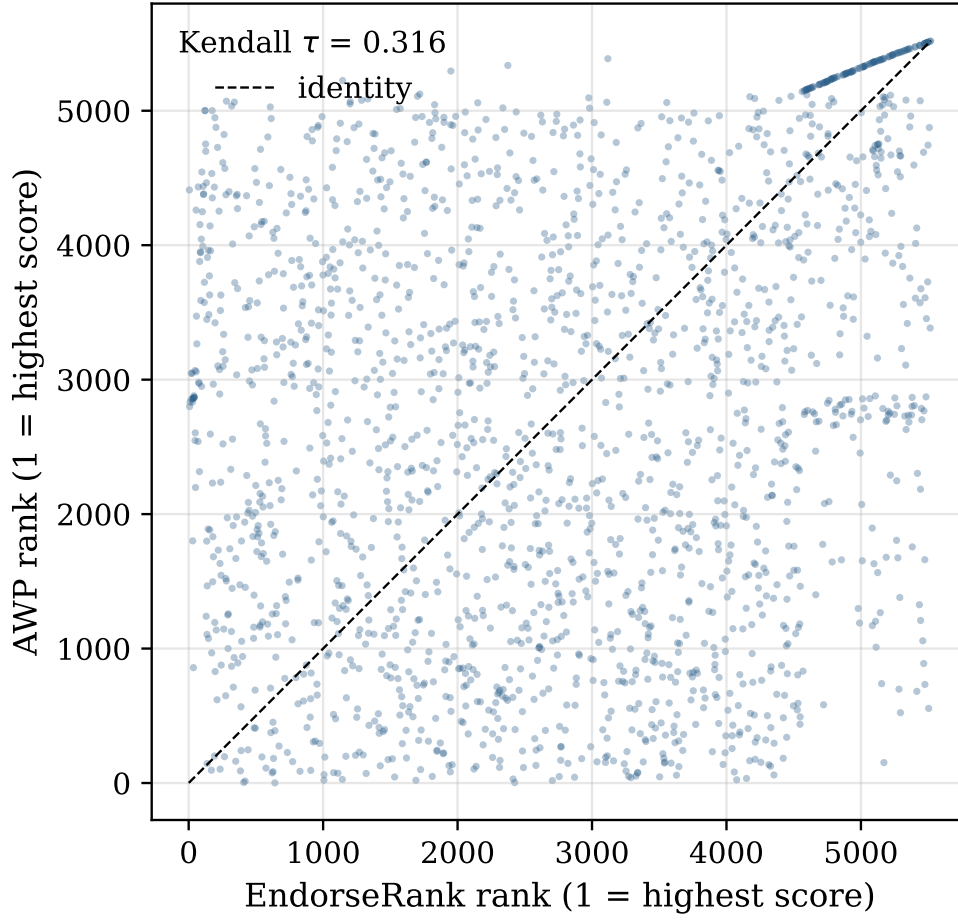


Figure 4.4: Rank–rank scatter of EndorseRank versus AWP on the matched cohort (random sub-sample of 2,000 wallets). Points on the dashed identity line would indicate identical orderings.

Table 4.8: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched alignment cohort ($n = 5,521$). The $d = 0.85$ runtimes are the same measurements as Table 4.1 (one timing session; each graph–damping pair is timed once).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.355	-0.031	0.333	0.531	0.594	4.420
0.85	0.355	-0.030	0.333	0.534	0.401	4.105
0.95	0.355	-0.030	0.333	0.536	0.427	5.420

damping operator; EndorseRank’s variation across the sweep is of the same order as its between-run spread in Table 4.1 and is not interpreted. The runtime ratio therefore depends on the damping setting even though the alignment conclusion does not. Figure 4.5 plots the family means.

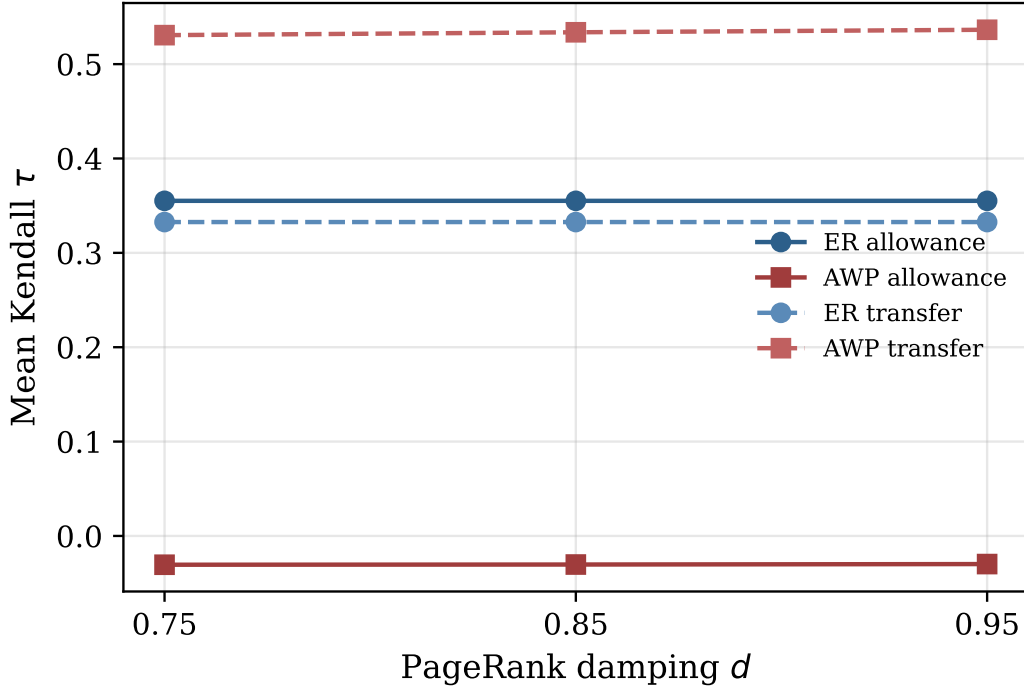


Figure 4.5: Damping sensitivity of family-mean Kendall τ for the allowance and transfer families, $d \in \{0.75, 0.85, 0.95\}$ (values of Table 4.8).

4.6.2 Top-token subgraph

Table 4.9: Robustness: mean Kendall τ on the top-20 ERC-20 token subgraph (matched cohort), next to the full-token value of Table 4.3 for the same family.

Method	Transfer		Allowance		Sybil	
	top-20	full	top-20	full	top-20	full
EndorseRank	0.331	0.333	0.355	0.355	0.234	0.233
AWP	0.532	0.534	-0.031	-0.030	0.283	0.286

Restricting both graphs to the twenty highest-volume ERC-20 tokens changes every family mean by at most a few thousandths. The alignment pattern is therefore not driven by long-tail tokens with

sparse or spam-like edges; the economically salient core of ERC-20 activity suffices to reproduce the EndorseRank–AWP contrast.

4.6.3 Sample-definition sensitivity

Table 4.10: Sensitivity of alignment to the evaluation set: mean Kendall τ and runtime at staged subsample sizes drawn from the expanded wallet pool ($N = 99,080$; seed `benchmark-tier2-v1`). Rows are evaluated on different wallet sets from the matched cohort, so τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. Runtimes are the same measurements as Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)	Speedup
10,000	0.343	0.242	0.821	0.904	0.070	0.854	12.20
50,000	0.330	0.232	0.828	0.915	0.334	2.706	8.10
99,080	0.327	0.230	0.828	0.916	0.401	4.105	10.24

Table 4.10 recomputes the transfer and allowance family means on the staged pool subsamples used for the runtime scaling. The coefficients are much larger than on the matched cohort, and AWP’s allowance coefficient turns positive. Neither change is evidence about the methods. Pool subsamples admit wallets that have no edges in one or both graphs; for such wallets the score and the proxy are both zero after alignment, and tied zeros at the bottom of both rankings raise Kendall τ mechanically (Section 3.12.4). The table is therefore reported as a sensitivity analysis of the sample definition, and it answers a narrower question: does the relative ordering of the two methods on each family survive a change of population? It does, with AWP ahead on transfer and EndorseRank on allowance at every stage, but the absolute values in the table must not be compared with those of Table 4.3, whose cohort excludes exactly the wallets that inflate them.

The damping and top-token checks support reading Claims C1–C3 as stable under parameter and token-mix perturbation. The sample-definition check shows that the comparative structure survives a change of population while the decimal values do not.

4.7 Coupled operator: same-window results (RQ5)

Section 3.6.3 defined the coupled operator C-PR, whose transition row at each wallet mixes the allowance and transfer layers with weight λ , and the endorsement-seeded walk S-PR, which keeps the transfer layer and restarts from the EndorseRank distribution. Table 4.11 places both next to the two single-layer methods on the three same-window families of Table 4.3; Table 4.12 tests the eight hybrid contrasts of Section 3.10 against the single-layer method that leads each family. The out-of-window test that decides RQ5 is in Section 4.8; this section reports where the hybrids sit on the checks that the single-layer methods were built to pass.

Table 4.11: Same-window family-mean Kendall τ with bootstrap intervals for the two single-layer methods, the coupled operator at three values of λ and the endorsement-seeded walk (matched cohort, $n = 5,521$). $\lambda = 1$ would reproduce EndorseRank and $\lambda = 0$ AWP on a shared node set. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.333 [0.314, 0.353]	0.355 [0.322, 0.384]	0.233 [0.214, 0.253]
AWP	0.534 [0.522, 0.547]	-0.030 [-0.045, -0.017]	0.286 [0.274, 0.300]
C-PR ($\lambda = 0.25$)	0.531 [0.520, 0.545]	-0.009 [-0.020, 0.004]	0.288 [0.276, 0.302]
C-PR ($\lambda = 0.5$)	0.527 [0.515, 0.540]	-0.001 [-0.013, 0.013]	0.290 [0.279, 0.304]
C-PR ($\lambda = 0.75$)	0.519 [0.507, 0.533]	0.006 [-0.007, 0.020]	0.292 [0.281, 0.306]
S-PR	0.494 [0.481, 0.508]	0.116 [0.097, 0.134]	0.220 [0.207, 0.235]

C-PR behaves like AWP with a small allowance correction. On the transfer family its coefficient falls from 0.531 at $\lambda = 0.25$ to 0.519 at $\lambda = 0.75$, all within a few thousandths of AWP’s 0.534; at the primary setting the paired difference to AWP is -0.007 with an interval that excludes zero but is narrower than the rounding of the table. On the allowance family C-PR is indistinguishable from zero at every λ , far below EndorseRank’s 0.355 ($\Delta\tau = -0.356$, interval $[-0.388, -0.323]$). The reason is structural. The transfer layer has about nine edges for every allowance edge and 30,791 nodes against 6,442 (Table 4.1), so most nodes of the union graph have no allowance out-edge at all. At such a node the mixing rule of Equation (3.6) uses the transfer row alone whatever the value of λ , and λ acts only at the minority of nodes that have out-edges in both layers. Once the walk has left

Table 4.12: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the single-layer method that leads each family (matched cohort, $n = 5,521$). Family-mean τ ; the same wallet resamples are used for a and b . The secondary criterion of Section 3.7 is met on a family when the interval includes 0 or lies above it (transfer, allowance) or lies above 0 (Sybil stability). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
C-PR transfer minus AWP transfer	0.527	0.534	-0.007	[-0.009, -0.005]	0.000
C-PR allowance minus EndorseRank allowance	-0.001	0.355	-0.356	[-0.388, -0.323]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.290	0.286	+0.004	[0.003, 0.006]	1.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.290	0.233	+0.057	[0.043, 0.069]	1.000
S-PR transfer minus AWP transfer	0.494	0.534	-0.040	[-0.049, -0.030]	0.000
S-PR allowance minus EndorseRank allowance	0.116	0.355	-0.239	[-0.263, -0.215]	0.000
S-PR sybil-stability minus AWP sybil-stability	0.220	0.286	-0.065	[-0.072, -0.058]	0.000
S-PR sybil-stability minus EndorseRank sybil-stability	0.220	0.233	-0.013	[-0.025, -0.003]	0.007

that minority it moves on transfer edges until it teleports. Raising λ to 0.75 moves the allowance coefficient by less than one hundredth. The convex mixture does not reach the allowance construct until λ is close to one, at which point it is EndorseRank.

The Sybil-adjusted stability family is the one place where C-PR exceeds both single-layer methods, as the secondary criterion of Section 3.7 asked: 0.290 against 0.286 for AWP and 0.233 for EndorseRank, with both paired intervals above zero ([0.003, 0.006] and [0.043, 0.069]). The increment over AWP is four thousandths. It is a difference in the statistical sense and not in any operational one.

S-PR is worse than the method it borrows from on every family. Seeding the transfer walk from the EndorseRank distribution lowers the transfer coefficient to 0.494 ($\Delta\tau = -0.040$ against AWP), lifts the allowance coefficient only to 0.116 ($\Delta\tau = -0.239$ against EndorseRank) and reduces Sybil-adjusted stability below both single-layer methods. A restart vector changes where the walk begins; on a graph of this density it does not change where the walk spends its time. That is the same-window form of the finding that Section 4.8 reaches out of window.

Against the secondary criterion, C-PR at $\lambda = 0.5$ passes two of its three parts and fails the third. Its transfer coefficient (0.527) lies inside AWP’s interval ([0.522, 0.547]); its Sybil-adjusted stability

exceeds both single-layer methods with intervals above zero; its allowance coefficient (-0.001) lies far outside EndorseRank’s interval ($[0.322, 0.384]$). The criterion is therefore not met. What the same-window tables show is that the mixture is dominated by the denser layer. Whether it carries any predictive content that the single layers lack is the question the holdout answers.

4.8 Temporal holdout (RQ4, RQ5)

The same-window checks cannot separate a method from its own construct. The holdout asks instead whether centrality measured up to a cut-off predicts behaviour after it. Scores are frozen at 28 February 2026 on the spender cohort and two labels are counted on each wallet between 1 March and 31 May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct), following Section 3.7. Table 4.13 reports every score computed at t_1 , including the two raw-degree baselines, against both labels; Table 4.14 reports the eight pre-registered paired contrasts.

Table 4.13: Temporal holdout on the spender cohort ($n = 1,335$): Kendall τ between scores frozen at 28 February 2026 and two labels counted on the wallet between 1 March 2026 and 31 May 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the wallet for the first time. CI = 95% percentile bootstrap over wallets (400 paired resamples). 222 wallets received at least one new pair and 175 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.711 [0.665, 0.759]	0.404 [0.348, 0.467]
Transfer in-degree at t_1 (baseline)	0.117 [0.049, 0.191]	0.472 [0.423, 0.520]
EndorseRank	0.479 [0.427, 0.529]	0.372 [0.325, 0.424]
AWP	0.044 [-0.011, 0.098]	0.270 [0.235, 0.310]
C-PR ($\lambda = 0.25$)	0.271 [0.231, 0.314]	0.266 [0.231, 0.306]
C-PR ($\lambda = 0.5$)	0.295 [0.252, 0.336]	0.277 [0.241, 0.318]
C-PR ($\lambda = 0.75$)	0.314 [0.271, 0.357]	0.287 [0.253, 0.330]
S-PR	0.114 [0.052, 0.164]	0.264 [0.229, 0.304]

Three findings follow, in the order of the contrasts.

First, the comparison between the single-layer methods on the endorsement label stands. EndorseR-

Table 4.14: Paired bootstrap differences on the spender holdout ($n = 1,335,400$ resamples shared by a and b). The first two rows give the increment of each PageRank over its own raw degree at t_1 ; the next two compare the single-layer methods on the other construct’s label; the last four are the pre-registered hybrid contrasts. Primary criterion (C-PR no worse than EndorseRank on new approval pairs and better than AWP on new transfer senders): not met.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank minus t1 in-approve degree	new approval pairs	0.479	0.711	-0.233	[-0.291, -0.183]	0.000
AWP minus t1 in-degree	new transfer senders	0.270	0.472	-0.201	[-0.230, -0.171]	0.000
EndorseRank minus AWP	new approval pairs	0.479	0.044	+0.435	[0.365, 0.511]	1.000
AWP minus EndorseRank	new transfer senders	0.270	0.372	-0.102	[-0.148, -0.051]	0.000
C-PR minus EndorseRank	new approval pairs	0.295	0.479	-0.184	[-0.213, -0.149]	0.000
C-PR minus AWP	new transfer senders	0.277	0.270	+0.007	[-0.012, 0.028]	0.735
S-PR minus EndorseRank	new approval pairs	0.114	0.479	-0.365	[-0.431, -0.300]	0.000
S-PR minus AWP	new transfer senders	0.264	0.270	-0.006	[-0.017, 0.006]	0.163

ank’s coefficient on new approval pairs ($\tau = 0.479$) is larger than its same-window allowance coefficient, and its paired difference to AWP ($\Delta\tau = +0.435$, interval $[0.365, 0.511]$) is the largest in the chapter; AWP’s own interval on this label includes zero. Because the labels post-date every scored event, this is out-of-window evidence that endorsement centrality carries information about future endorsement that transfer centrality does not. On the flow label the ordering is the same but smaller: EndorseRank (0.372) exceeds AWP (0.270) with an interval that excludes zero ($[-0.148, -0.051]$ for AWP minus EndorseRank). On this cohort an approval tends to precede a transfer from the same counterparty, so the wallets that gain new owners also gain new senders.

Second, neither PageRank improves on its own raw degree at t_1 , and RQ4’s second clause is therefore answered in the negative. In-approve degree at t_1 predicts new approval pairs with $\tau = 0.711$ against EndorseRank’s 0.479; the paired increment is $\Delta\tau = -0.233$ with interval $[-0.291, -0.183]$. Transfer in-degree at t_1 predicts new transfer senders with $\tau = 0.472$ against AWP’s 0.270 ($\Delta\tau = -0.201$, interval $[-0.230, -0.171]$). Both intervals lie well below zero. The count of new counterparties in a three-month window grows with the number of existing counterparties, and the local count captures that directly; PageRank replaces the count with a weighted, propagated quantity in which owner-side centrality and allowance value enter as well, and on this label those additional terms act as noise. The out-of-window predictive content of EndorseRank is

contained in, and diluted from, the spender’s t_1 in-approve degree. The same statement holds for AWP and the transfer in-degree. Whatever either PageRank adds over its degree must therefore be sought in same-window structure (Sections 4.3 and 4.5), not in prediction of the construct’s own future growth.

Third, the coupled operator does not meet the pre-registered primary rule, on either part. C-PR at $\lambda = 0.5$ is worse than EndorseRank on new approval pairs ($\Delta\tau = -0.184$, interval $[-0.213, -0.149]$) and indistinguishable from AWP on new transfer senders ($\Delta\tau = +0.007$, interval $[-0.012, 0.028]$). The three values of λ order monotonically on both labels: τ on new approval pairs rises from 0.271 ($\lambda = 0.25$) through 0.295 to 0.314 ($\lambda = 0.75$), and on new transfer senders from 0.266 through 0.277 to 0.287. More allowance weight helps on both labels and the transfer layer adds nothing out of window for this cohort; the coupled score interpolates between the two single-layer scores rather than exceeding either. The seeded ablation shows why. S-PR matches AWP on the flow label ($\Delta\tau = -0.006$, interval $[-0.017, 0.006]$) and falls far below EndorseRank on the endorsement label ($\Delta\tau = -0.365$): once the transfer structure governs propagation, an endorsement-derived restart distribution changes little, so the value of the allowance edge lies in propagating over it, not in seeding from it.

Two limits apply to the whole table. Each label is the same construct in the future tense (new approvals, new senders), so nothing here speaks to credit risk or loss. And both labels are sparse: 222 of 1,335 wallets received a new approval pair and 175 a new sender, so every coefficient is driven by the ordering among that minority.

Other candidate labels computed for the holdout are not reported as support. Revoke counts and the drain heuristic correlated positively with EndorseRank, which has the wrong sign for an endorsement reading (a heavily endorsed spender is also a frequently revoked one), and Aave liquidation events on this spender set were empty. The trading-success comparison of the research proposal was withdrawn for the reason given in Section 2.8: the outcome-native reference it relied on is built from the same GMX events as the proxies it would be validated against.

4.9 Primary claims (C1–C5)

The results of this chapter support the following claims, mapped to hypotheses and research questions in Table 1.2.

- C1. On identical wallet sets EndorseRank solves PageRank in a fraction of AWP’s wall-clock time, with proportionally lower memory and fewer iterations, because its latest-allowance graph has about one ninth of the transfer graph’s edges (Table 4.1). The ratio is stable across in-cohort and pool stages whose edge counts vary by a factor of five to seven (Tables 4.2 and 8.1); the pool stages do not test growth in edge count beyond the cohort graph. The claim is specific to EndorseRank: the coupled operator walks on both edge sets and costs at least as much as AWP.
- C2. Each method recovers its intended construct on same-window local statistics, with bootstrap intervals well away from zero (Table 4.3). These are construct checks, not external validation.
- C3. The winners are construct-specific: AWP exceeds EndorseRank on the transfer family and EndorseRank exceeds AWP on the allowance family, with paired-difference intervals that exclude zero in opposite directions (Table 4.4); the two score vectors agree only moderately (Table 4.3). EndorseRank’s allowance lead over its own transfer alignment is not statistically established.
- C4. The efficiency and construct results combine into a trade-off. EndorseRank is the cheaper method and the only one that aligns with authorisation checks, and on the spender holdout it predicts future new approvals where AWP does not (Table 4.13). The out-of-window claim is bounded by Table 4.14: the spender’s raw in-approve degree at t_1 predicts the same label better than EndorseRank, and the transfer in-degree predicts new senders better than AWP. AWP remains the stronger method for transfer-hub detection and longitudinal stability.
- C5. Coupling the two edge types in one walk does not improve on either single-layer method. At every λ tested C-PR lies between EndorseRank and AWP on both out-of-window labels, the pre-registered primary rule is not met, and the seeded ablation reproduces AWP (Tables 4.13

and 4.14). The same-window position of the hybrids is reported in Section 4.7. C5 is a bounded negative result: on this chain and cohort the two edge types do not combine profitably under a per-node convex mixture with fixed λ .

RQ1 is answered by the inter-method agreement and Figure 4.4; RQ2 by the family tables and difference tests; RQ3 by the benchmark tables; RQ4 and RQ5 by the holdout. The claims do not retire AWP, and they do not promote EndorseRank beyond what the intervals allow: EndorseRank is cheaper, aligns with allowance checks, diverges moderately from AWP, predicts later new approvals where AWP does not, and is itself out-predicted by the raw degree it smooths.

Chapter 5 interprets these findings against the research questions and objectives of Chapter 1, draws out their theoretical, practical and societal implications, and examines the threats to validity and limitations that bound them.